Code Kernel '26 • Problem Statement • In collaboration with Canvas Classes

Adaptive Learning for Classes 6 to 10

Decode Round | 21 to 24 October 2026

THE PROBLEM

One pace does not fit every learner

Every classroom has students learning at different speeds. Some have already understood a topic while others are still stuck on the basics behind it. Yet most students get the same lessons, the same questions and the same pace.



Fast learners get bored

Some have already understood the topic and are kept at the same pace.

Struggling students fall behind

Others are still stuck on the basics behind it.

Nobody really knows

No clear view of what each student has understood.

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WHY ONE PACE FAILS

The gap grows quietly

In most classrooms, the whole class moves through the syllabus at one speed. Students who already understand a topic wait, while students who missed an earlier idea try to build on a gap.

Tests usually come at the end of a chapter, so by the time a gap shows up, the class has already moved on. Teachers want to help every student, but in a full classroom it is hard to see who needs what.



STUDENTS AGED 11 TO 16

What students in Classes 6 to 10 face

Concepts build up

A basic missed in Class 6 or 7 makes the topics built on it harder later.

Rising pressure

Closer to Class 10 and board exams, there is less time to fix old gaps.

Many languages

Many students learn in a language different from the one spoken at home.

Outside school

Tuition, coaching and apps add hours of study, but not always more understanding.

Uneven access

Some students have their own device and fast internet. Many share a family phone or have limited data, and some have neither.

THE INDIAN CONTEXT

Built for India

Most adaptive learning apps today are built for other markets. Their algorithms aren't designed for Indian classrooms: our school boards and syllabi, large class sizes, many languages, and students with very different access to devices and internet.

THE ROLE OF THE TEACHER

Empower the teacher

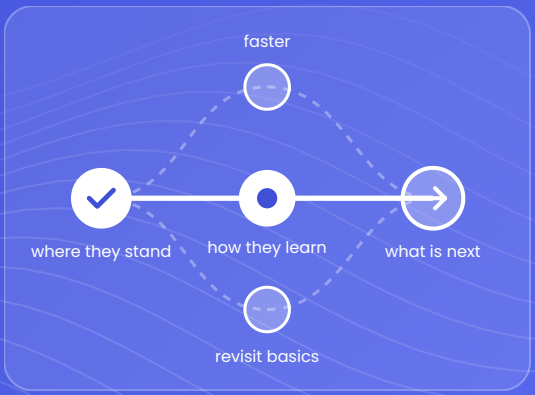
School teachers are increasingly being sidelined by coaching centres and AI tools. Your solution should put teachers back at the centre, helping them understand each student, personalise their teaching and guide learning, instead of replacing them.



YOUR CHALLENGE

Make learning adapt to every student

Design a solution that adapts learning to each student in Classes 6 to 10. It should understand where a student stands and change what they learn next, how they learn it, or how fast they go.



MUST HAVE

Your solution must be built for Indian students and Indian classrooms. Solutions that simply copy existing global apps will miss the point.

GUIDING QUESTION

How can your solution empower the school teacher?

THREE PARTS OF STUDYING

Learn, Solve, Test

Adaptive learning is not just about practice questions. Studying has three parts: learning a concept, solving problems and testing what you know. **Your solution should adapt across all three.**

01	02	03
Learn	**Solve**	**Test**
Learning a concept	Solving problems	Testing what you know

QUALITIES OF A STRONG SOLUTION

What we're looking for

Understands where each student stands
It finds out what a student knows and where they are stuck.

Adapts learning, solving and testing
It changes what comes next, how and how fast, in all three parts.

Fits Indian classrooms
Boards, languages, class sizes, and real device and internet access.

Empowers teachers
It helps teachers understand and guide each student, not replace them.

Practical and usable
It is simple and engaging enough for students aged 11 to 16 to actually use.

BEFORE YOU START

Key questions to think about

01 How will you know what a student has understood?

02 How will your solution adapt when a student struggles or moves ahead?

03 How will it work in a typical Indian school?

04 How does it help the teacher, not replace them?

05 Who uses it, and how often?

OPEN TO YOUR INNOVATION

You decide

The subject, the users, the approach and the technology are up to you. It could be for students, teachers, parents or all of them. Think beyond quizzes, and surprise us.

Subject · Users · Approach · Technology

Guidance, not a checklist. The points in this document are here to guide your thinking. You do not have to tick every box.

DELIVERABLES

What to submit for Decode

01 · The problem

Your understanding of the problem and who you're solving it for.

02 · Your idea

Your idea and how it adapts to each student.

03 · The plan

Your planned approach and tech stack.

THREE ROUNDS, ONE IDEA

The journey

Decode
21 to 24 Oct
CURRENT

Build
25 to 29 Oct
UP NEXT

Prove
30 to 31 Oct
FINAL

Details for the Build and Prove rounds will be shared as the event goes on.